

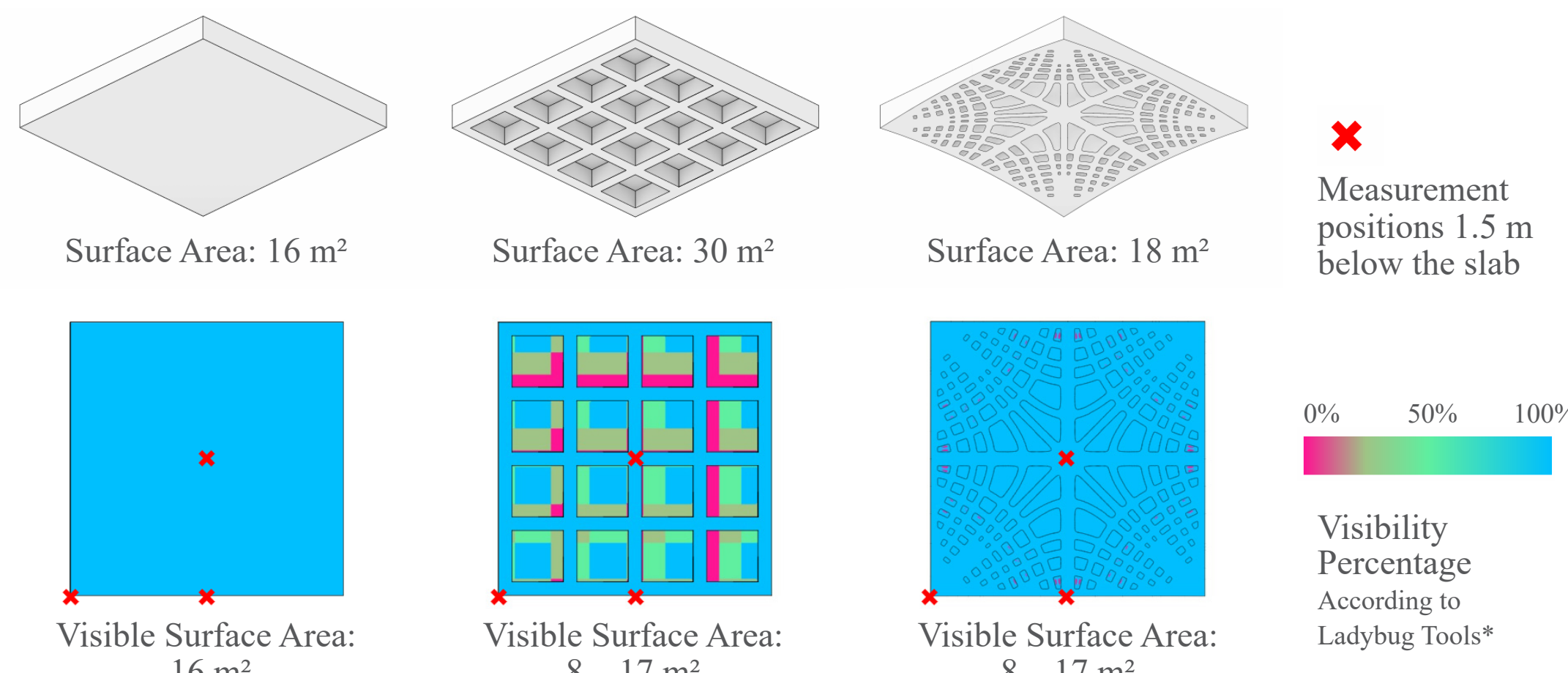
Vault House

Yiran Li, Erin Wai In Wong, Sebastián Canovas Quintana

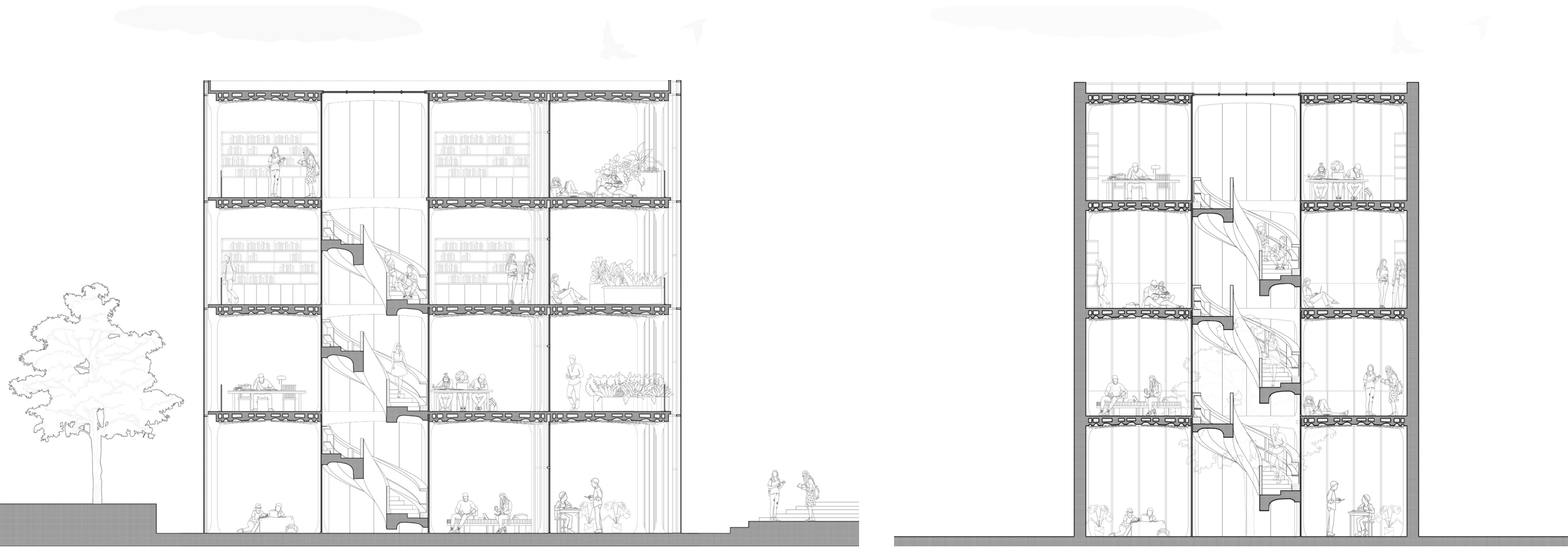


Project Description:

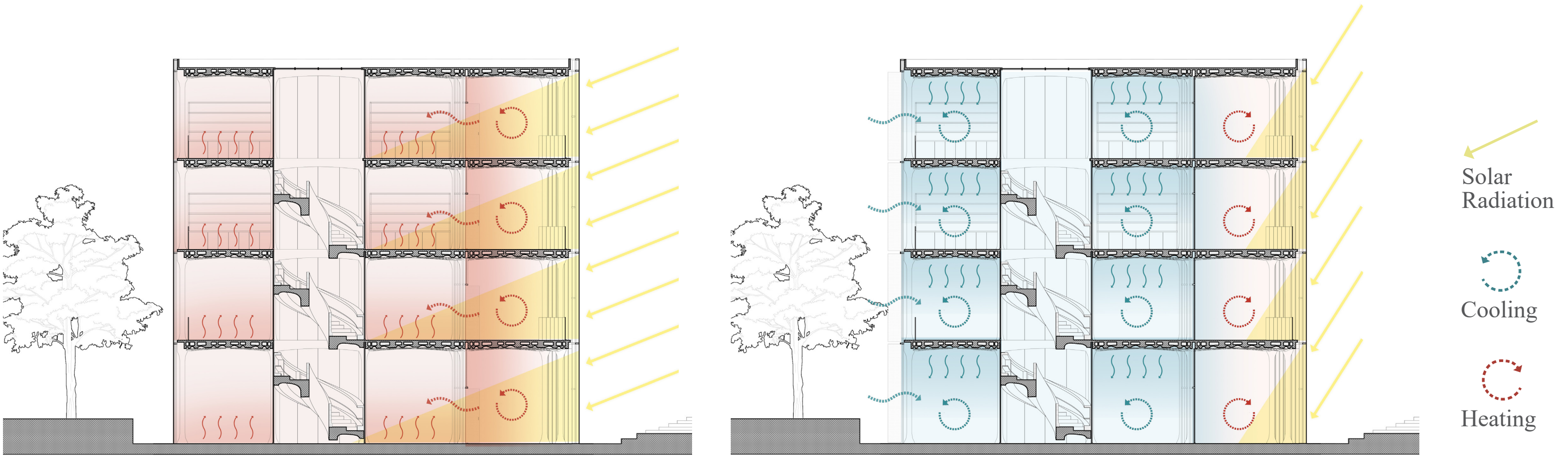
We would like to envision a modern library space due to the large number of students in Heidelberg, which embodies a fusion of learning, relaxation, and community. Apart from the design strategy, we have also focused on climatic sustainability. Our design features the a structurally optimized slab system, which reduces material and at the same time increases its surface area and its view factor to improve its performance as a thermal radiator and convector. The transparent facade of the building emphasizes the lightness of the system and allows the slabs to be seen from the square reminding of the coffered vaults of the historic Altes Hallenbad while showing exciting possibilities for the future of digital fabrication in the construction industry.



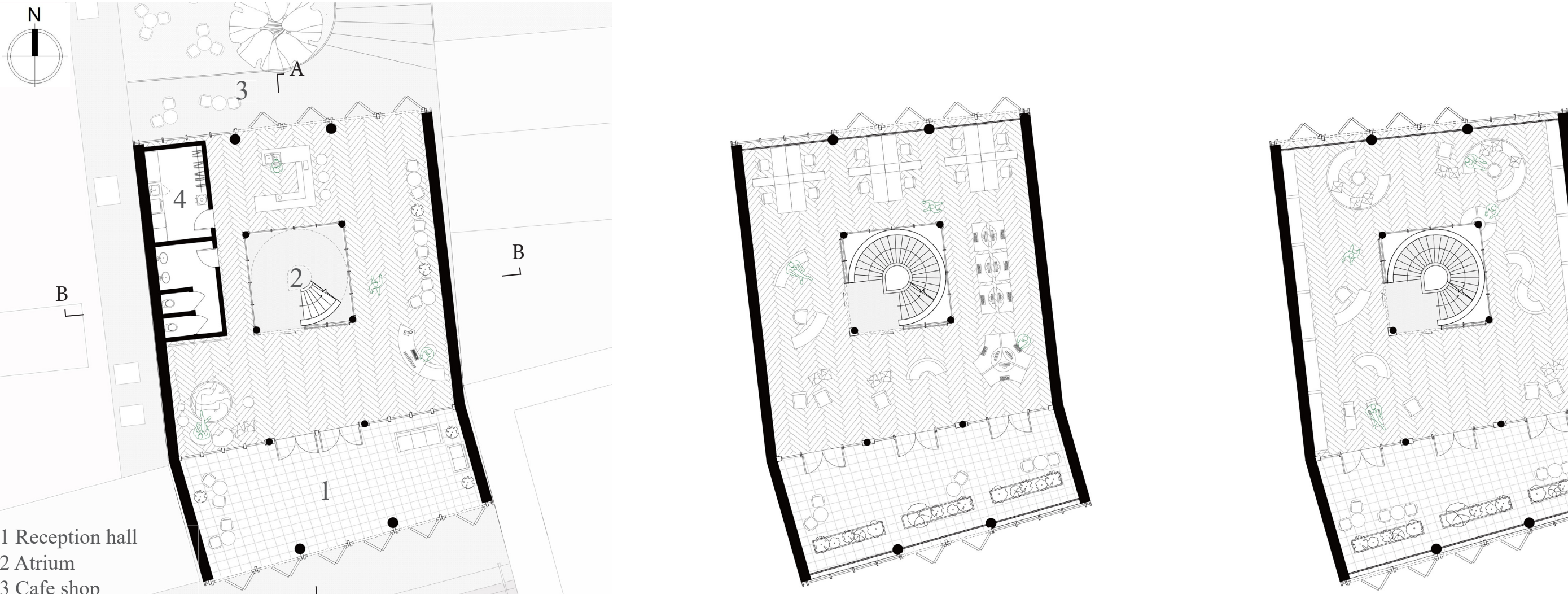
Visibility Percentage Comparison



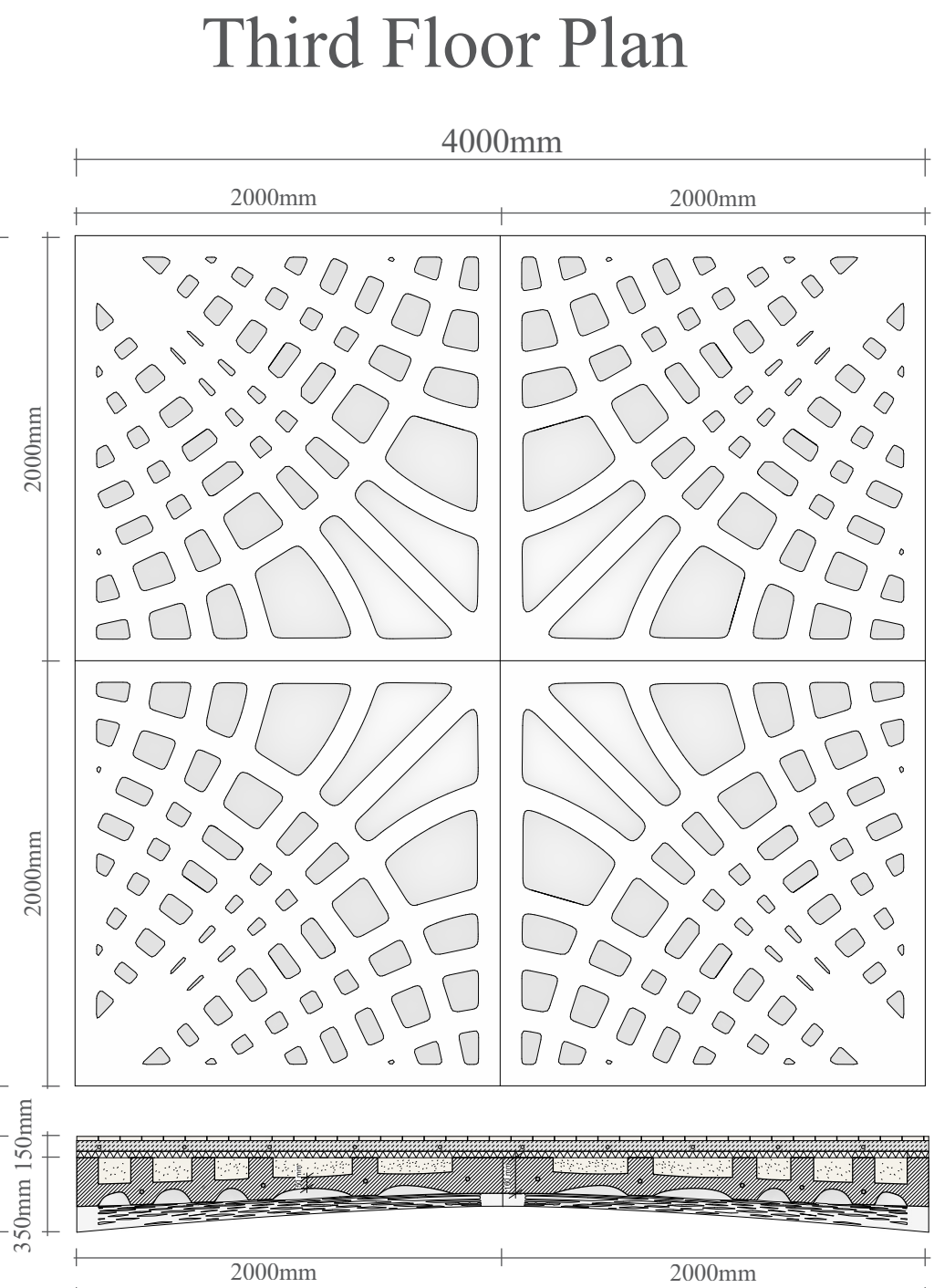
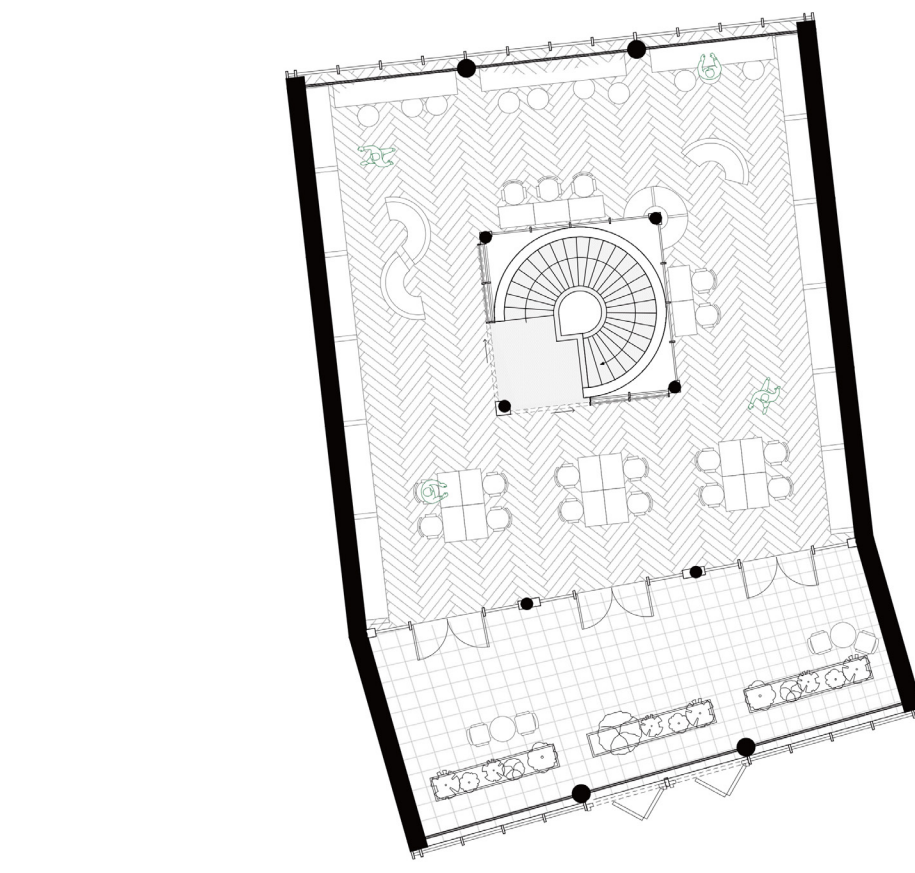
Section A-A Section B-B



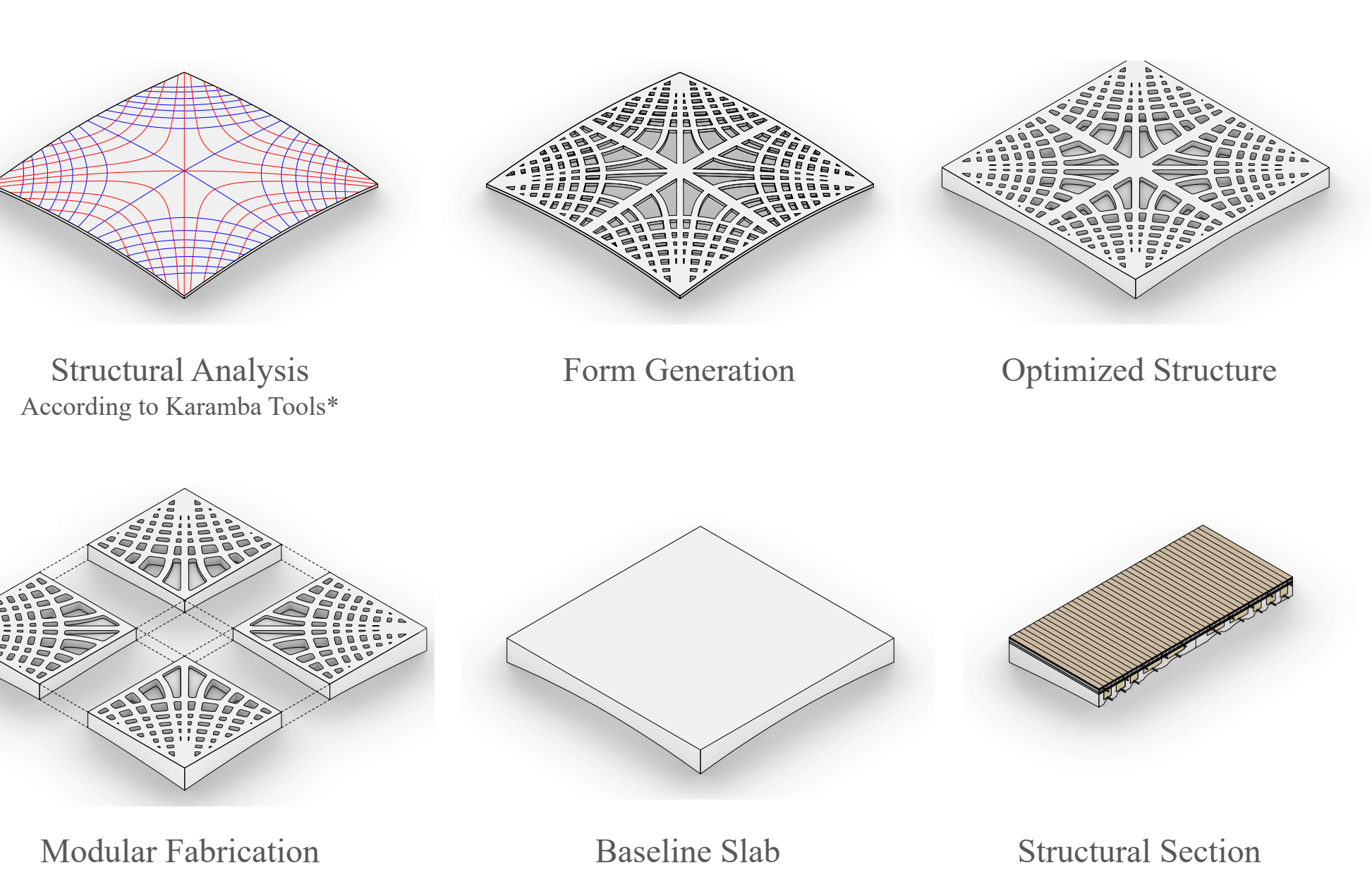
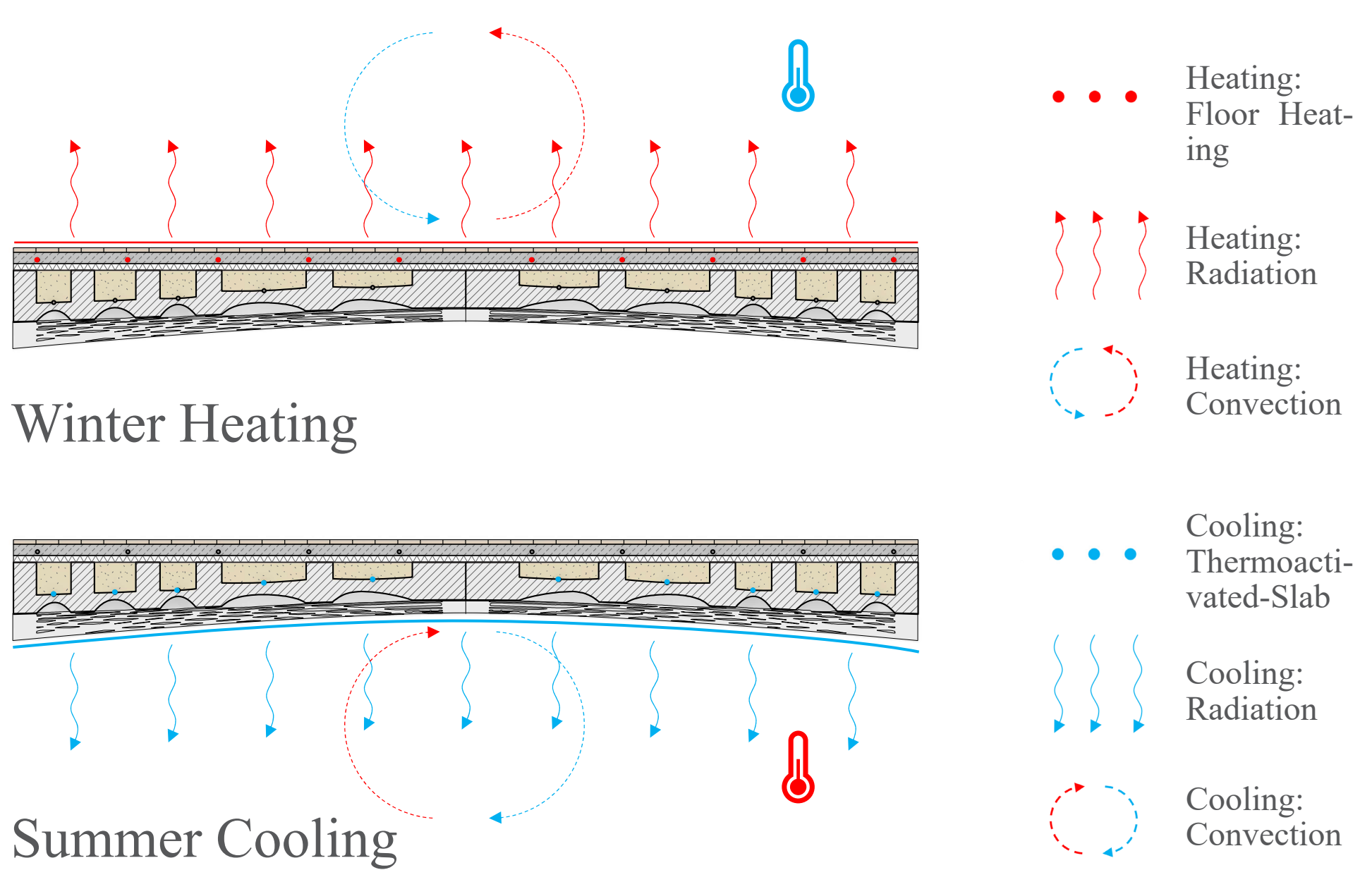
Climate Concept Winter: 17° (Min. Incidence angle) Summer: 64° (Max. Incidence angle)



Elevation



Slab details



Development of the Slabs

